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The Human Impacts of Artificial Intelligence Data Centers – UAB Institute for Human Rights Blog

10–12 minutes

This summer, I worked with a few different advocacy organizations during Louisiana’s 2025 Congressional Session. The amount of policy issues flying around was mind-spinning, but a constant murmur about the new Meta data center popping up in Richland Parish always seemed to pierce through the chaos. I couldn’t help but think, “Of all the state issues we could be debating, what could be so provocative about a data center?”

Data centers are nothing new; ever since the birth of the Internet, they have been used for the large-scale computing that comes with ever-advancing technology. With the rapid expansion of generative AI, our country is seeing more and more of these processing centers pop up, especially in rural areas.

Governments, researchers, and communities alike have been forced to face the glaring reality that comes with the construction and maintenance of new AI data centers: where there are new data centers, there are human lives directly impacted by their creation. Debate on whether these effects are a net positive or negative to these communities has prompted closer examination on the human impact of data centers. Only through a thorough

analysis of this ongoing research can we determine the nature and scope of these impacts and explore proper policy responses.



Source: Adobe Express, Sepia100, #566722487

WATER

We rely on water; it's as simple as that. We need water to drink, bathe, flush the toilet, wash our hands and dishes, and water our crops; it's a necessity to life, and an officially recognized human right. As much as we need water, data centers are even thirstier. It takes a lot of water to cool down all of the computing that takes place in these buildings. In 2021, just one of Google's data centers in Oregon used up [355 million gallons of water](#). In 2023, all of Meta's data centers worldwide guzzled around 1.4 billion gallons of water. Where is this water coming from? Of Meta's 1.4 billion gallons, about 672 million gallons came from local water sources. The extraction process is permanent, meaning data centers deplete millions of gallons of water from communities' local water supply yearly, and with the industry's rapid expansion, its water consumption will only grow. Some residents living nearby these new data centers, such as [Beverley Morris in Mansfield, Georgia](#), believe that these centers are draining wells and aquifers, leaving

locals without drinkable or fully functional running water in their homes. For communities in the Southwest, this could pose an especially [pressing threat](#) during droughts as the scarce water supply is divided between industrial and civilian use.

[Landon Marston](#), a professor in environmental and water resources engineering at Virginia Tech University, points out that since companies like Meta and Google tend to choose areas outside of cities to construct these data centers, the surge in water demand could also necessitate water infrastructure updates, the costs of which could fall partly on local ratepayers.

ENERGY

AI data centers require tons of energy. We're talking [200 trillion watts an hour](#), and that was only in 2016. The power usage of these data centers is projected to rise to nearly 2967 trillion watts an hour by 2030. The previously flatlined demand for electricity has been [increasing nationally since 2023](#) partly due to the energy-intensive operations of growing data centers. The majority of data centers' energy relies on fossil fuels and power plants, putting pressure on local energy grids. This increased pressure poses the threat of more frequent, long-lasting, and expensive blackouts for the communities surrounding these energy-hungry data centers.

More pressure on the grid naturally means more pressure to update the grid. Local belief and research alike contend that the cost of these grid updates, as well as the price tag of the extra energy demand, will show up in locals' energy bills. [A Harvard study](#) provides evidence that under-the-table agreements between

utilities and Big Tech consumers could be partly responsible for increased rates on everyday residents' bills. Additionally, in places like [Louisiana](#), the combination of prolonged need for air conditioning and damage to energy infrastructure due to storms drive energy bills up as it is; the intense energy demands of the new data center will serve only to exacerbate the steep cost of energy and amenities in nearby homes and businesses. Utilities are essential to decent quality of life and even employment, tying their accessibility directly to human rights.



Source: Adobe Express, Anna, #529027855

PUBLIC HEALTH

Since AI data centers rely heavily on the fossil-fuel energy of power plants, they run the risk of increasing local pollution and threatening public health in already vulnerable rural locations. AI centers, on top of their energy use from the grid, also employ backup generators in case of grid failure; these diesel generators can release [200 to 600 times more nitrous oxides](#) (NO_x) than a natural gas plant while producing the same amount of energy. [NO_x pollution](#) can cause irritation in the eyes, throat, and nose, as well

as more severe cases of respiratory infection, reduced metabolism, and even death. [According to the Institute of Electrical and Electronics Engineers, IEEE](#), data centers caused about \$6 billion in public health damages due to this type of air pollution in 2023. That being said, location matters. Often, these data centers choose rural areas, and in cases like that of Bessemer, Alabama, these areas are often home to a large Black population. Black Americans already suffer disproportionately from air pollution and other environmental injustices; in fact, low-income Black Americans have the [highest mortality rate](#) due to fine particulate matter air pollution. The emergence of data centers in rural Black communities only serves to [exacerbate this phenomenon](#). This can be directly traced to industrial [zoning policies](#), which often result in the sacrifice of poor, rural, often Black areas to attract business and wealth to cities. The result? Higher rates of asthma, respiratory issues, even pollution-related death, and a direct violation of the human right to clean air.



Source: Adobe Express, Jaroslav Pachý Sr., #175217425

ECONOMY

While industrial zoning and property value are the most important location factors, choosing a lower income, rural area also poses possible economic advantage for the communities. The construction of processing centers can require [thousands of workers](#), offering steady employment opportunities for locals. After construction, companies like Meta, Google, and Microsoft will have to hire employees to keep their data centers managed and running properly, another new job opening for those in the surrounding area. [Some locals have expressed excitement](#) over the new economic growth data centers will bring, especially in areas with dwindling industries like coal and timber. Working in data centers is an attractive alternative to the low-paying, dangerous agricultural jobs some of these areas rely on. Others have raised concerns that while many jobs will certainly appear during the construction period of the centers, employment opportunities from data centers seem to [fall off afterwards](#). Depending on the size, each data center building could operate with as little as fifty employees, [according to Microsoft](#). Larger ones like the one developing in Louisiana are required to [employ 500 locals](#), but even that opportunity seems small to some residents in comparison to the harm the center could bring to their community. Members of communities impacted by the development of data centers have also expressed [concerns about land usage](#), pointing out that the extensive land taken up by these new data centers had potential to be used for farming or other less health-damaging economic development. The right to employment good working conditions are outlined directly in the Universal Declaration of Human Rights, and these economic impacts could very well jeopardize them for

those living in surrounding areas.

What Now?

Artificial Intelligence isn't going away; in fact, we can expect its rapid expansion in the coming years, including the construction of dozens of new data centers. Behind AI's captivating technologies, there are human lives impacted by the processes it takes to power its functions. Considering the damage data centers can do to local resources, it certainly seems like measures need to be taken to ensure the escalating growth of AI doesn't come at the expense of communities, especially those that already face disadvantage. First and foremost, companies establishing these centers should focus on [using renewable energy](#) for much of their power, thereby decreasing their environmental impact on local communities. In addition, companies should [adopt initiatives](#) to maintain the local water supply's integrity, recycle water when possible, and ultimately, improve the efficiency of their computing to save resources like water and electricity. Local governments must ensure that the [price of increased pressure on electricity and water](#) infrastructure does not end up on ratepayers' bills; this means more transparency from large companies and their agreements with local utility providers and governments regarding the construction and maintenance of these centers and the impacts on local residents' well-being. These centers, if built sustainably and with people in mind, could ultimately have a positive impact on industry and economy within these communities. The development of data centers must not concentrate solely on maximum profit and computing power but also on the adverse effects the center has on utility bills, air quality, water demands, the power grid, and public

health as a whole.

So, really, it's no wonder advocates, lobbyists, and policymakers couldn't stop talking about Richland Parish's new data center. It's nearly as big as Manhattan, and its effects on the surrounding community may end up being just as sizable.